

# CSCI E-102 Econometrics and Causal Inference with R

Harvard Extension School

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Lecture 8

## 1 Joint Hypothesis

- F-test
  - Homoskedasticity-only F-statistic
  - Heteroskedasticity-robust F-statistic
- Joint Hypotheses: Bonferonni Correction

## 2 Model Specification

- Control Variables and Conditional Mean Independence
- External Validity
- Internal Validity

## 3 Regression with Panel Data

- Terminology
- Entity Fixed Effects
- Time Fixed Effects
- Both Entity and Time Fixed Effects

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# Joint Hypotheses: F-test

## Example:

Let's consider the following econometric model:

$$\text{ahe}_i = \beta_0 + \beta_1 \text{yrseeduc}_i + \beta_2 \text{age}_i + u_i,$$

where

$\text{ahe}_i$  = Average Hourly Earnings in current U.S. dollars;

$\text{yrseeduc}_i$  = Years of Education;

$\text{age}_i$  = Age (in Years).

Assume, we want to test

$$H_0 : \beta_1 = 0 \text{ and } \beta_2 = 0 \text{ vs. } H_1 : \beta_j \neq 0 \text{ for some } j \in \{1, 2\}.$$

# Joint Hypotheses: F-test

## Example (continued):

Under homoskedasticity assumption, using `lm()` function we get:

```
> fit <- lm(ahe ~ yrseduc + age, data = df)
> summary(fit)
```

Residuals:

| Min     | 1Q     | Median | 3Q    | Max    |
|---------|--------|--------|-------|--------|
| -36.260 | -7.629 | -2.257 | 5.121 | 97.591 |

Coefficients:

|             | Estimate   | Std. Error | t value | Pr(> t )   |
|-------------|------------|------------|---------|------------|
| (Intercept) | -18.443636 | 0.316070   | -58.35  | <2e-16 *** |
| yrseduc     | 2.340936   | 0.018881   | 123.98  | <2e-16 *** |
| age         | 0.189979   | 0.004388   | 43.30   | <2e-16 *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.01 on 63192 degrees of freedom

Multiple R-squared: 0.2202, Adjusted R-squared: 0.2202

F-statistic: 8922 on 2 and 63192 DF, p-value: < 2.2e-16

# Joint Hypotheses: F-test

Example (continued):

where the homoskedasticity-only F-statistic is computed as follows:

$$F = \frac{(\text{SSR}_{\text{restricted}} - \text{SSR})/q}{\text{SSR}/(n - k - 1)}.$$

Here,

- $q = 2$  is the number of restrictions
- $(k + 1) = 3$  is the number of parameters
- $n = 63195$  is the number of observations
- SSR is the sum of squared residuals
- $\text{SSR}_{\text{restricted}}$  is the sum of squared residuals under  $H_0 : \beta_1 = 0$  and  $\beta_2 = 0$

Note:

- ✓ If errors are homoskedastic and normal,  $F \sim F_{q, n-k-1}$ .
- ✓ If errors are homoskedastic (not necessarily normal),  $F \overset{A}{\sim} F_{q, \infty}$  as  $n \rightarrow \infty$ .

# Joint Hypotheses: F-test

Example (continued):

What if the errors are heteroskedastic? We need to use the heteroskedasticity-robust standard errors to compute F-statistics.

Recall:

Let  $\mathbf{R}$  be  $q \times (k + 1)$  matrix. Assume we want to test

$$H_0 : \mathbf{R}\boldsymbol{\beta} = \mathbf{r} \text{ vs. } H_1 : \mathbf{R}\boldsymbol{\beta} \neq \mathbf{r}$$

for some  $q \times 1$  vector  $\mathbf{r}$ , then one can use the following TS:

$$\text{TS} = (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})^\top [\mathbf{R}\hat{\boldsymbol{\Sigma}}_{\hat{\boldsymbol{\beta}}}\mathbf{R}^\top]^{-1} (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})/q,$$

which approximately follows F-distribution for large  $n$ :

$$\text{TS} \xrightarrow{D} F_{q,\infty}.$$

# Joint Hypotheses: F-test

Example (continued):

$TS = (\mathbf{R}\hat{\beta} - \mathbf{r})^T [\mathbf{R}\hat{\Sigma}_{\hat{\beta}}\mathbf{R}^T]^{-1} (\mathbf{R}\hat{\beta} - \mathbf{r}) / q$  can be obtained in two steps:

- 1 The variance-covariance matrix  $\Sigma_{\hat{\beta}}$  of  $\hat{\beta} = (\hat{\beta}_0 \ \hat{\beta}_1 \ \hat{\beta}_2)^T$  is estimated as:

```
> Sigma = vcovHC(fit, type = "HC1")
> Sigma
```

|             | (Intercept)   | yrseduc       | age           |
|-------------|---------------|---------------|---------------|
| (Intercept) | 0.1092203979  | -6.053782e-03 | -7.189501e-04 |
| yrseduc     | -0.0060537823 | 4.544684e-04  | 3.384593e-06  |
| age         | -0.0007189501 | 3.384593e-06  | 1.700859e-05  |

- 2 The heteroskedasticity-robust F-statistic is computed as follows:

```
> R = matrix(c(0,1,0,0,0,1), nrow = 2, ncol = 3, byrow = TRUE)
> r = matrix(c(0,0), nrow = 2, ncol = 1, byrow = FALSE)
> q = nrow(r)
> b = fit$coefficients
> TS = t(R%*%b-r) %*% inv(R %*% Sigma %*% t(R)) %*% (R%*%b-r) / q
> TS
```

```
      [,1]
[1,] 6905.501
```

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# Joint Hypotheses: Bonferonni Correction

Claim (Bonferonni's Inequality)

Given  $q$  events  $A_1, A_2, \dots, A_q$ ,

$$P(A_1 \cup A_2 \cup \dots \cup A_q) \leq P(A_1) + P(A_2) + \dots P(A_q).$$

# Joint Hypotheses: Bonferonni Correction

## Claim (Bonferonni's Inequality)

Given  $q$  events  $A_1, A_2, \dots, A_q$ ,

$$P(A_1 \cup A_2 \cup \dots \cup A_q) \leq P(A_1) + P(A_2) + \dots P(A_q).$$

## Corollary (Bonferonni Correction)

Suppose we want to test a family  $\{H_1, H_2, \dots, H_q\}$  of  $q$  hypotheses, say,

$$H_0 : \underbrace{\beta_1 = 0}_{H_1}, \underbrace{\beta_2 = 0}_{H_2}, \dots, \underbrace{\beta_q = 0}_{H_q} \quad \text{vs.} \quad H_a : \text{at least one hypothesis is false}$$

simultaneously at significance level  $\alpha$ .

Here, we should run the F-test, but if data is not available, one can use the Bonferonni's inequality:

$$P(\{\text{reject } H_1\} \cup \dots \cup \{\text{reject } H_q\}) \leq P(\text{reject } H_1) + \dots + P(\text{reject } H_q)$$

and test each individual hypothesis at a significance level of  $\alpha/q$ .

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# Control Variables and Conditional Mean Independence

## Recall:

Omitted variable bias occurs if the following both conditions hold:

- 1 Omitted variable causes the response  $Y_i$ .
- 2 Omitted variable is correlated with one of the regressors.

If there is an omitted variable, which  $\beta$  coefficients will be biased?

# Control Variables and Conditional Mean Independence

Def.

We say  $u_i$  and  $X_{i1}$  are *conditionally mean independent*, given  $X_{i2}$ , if:

$$E[u_i | X_{i1}, X_{i2}] = E[u_i | X_{i2}].$$

Claim:

Assume the following Multivariate Linear Regression Econometric model,

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 \underbrace{X_{i2}}_{\text{control}} + u_i,$$

where  $u_i$  and  $X_{i1}$  are conditionally mean independent, given  $X_{i2}$ .

Then the OLS estimator provides an unbiased estimate of  $\beta_1$  (i.e. an unbiased estimate of the causal effect).

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# External Validity

## Def.

The analysis is said to have *external validity* if its inferences and conclusions can be generalized from the *population and settings studied* to other populations and settings. *Population studied* refers to population of entities from which the sample was drawn.

## Threats to External Validity:

- Differences between the population studied and the population of interest.
- Differences in settings such as laws, culture, institutional environment, etc.

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# Internal Validity

## Def.

A statistical analysis is said to have *internal validity* if the statistical inferences about causal effects are valid for the population being studied.

## Threats to Internal Validity:

- The estimator of the causal effect should be unbiased and consistent:
  - ▷ There should be no omitted variable bias - need to include omitted variables, if available; may use panel data to control for unobserved omitted variables; may find appropriate *instrumental variables*.
  - ▷ The functional form should be specified correctly, otherwise the estimator may be biased.
  - ▷ The estimator may be biased (towards zero) because of the *measurement errors*.
  - ▷ Non-randomly missing data may introduce bias - *sample selection bias*.
- Correctly estimated standard error should be used so hypothesis tests and confidence intervals have desired level  $\alpha$ .
- Simultaneous causality - need to design a randomized controlled experiment or use instrumental variables.

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# Regression with Panel Data

Terminology: *Panel data* (or *longitudinal data*) consists of observations on the same  $n$  entities at two or more time periods  $T$ :

$$(X_{it}, Y_{it}), \quad i = 1, 2, \dots, n, \quad \text{and} \quad t = 1, 2, \dots, T,$$

where

$i$  refers to entity and  $t$  refers to time at it is observed.

If some observations are missing the panel is called *unbalanced panel*, otherwise - *balanced panel*.

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# Regression with Panel Data: Entity Fixed Effects Regression

Let's assume that

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 Z_i + u_{it},$$

where  $Z_i$  is an unobserved variable that depends on  $i$  but does not change over time  $t$ . Therefore, we can rewrite

$$Y_{it} = \alpha_i + \beta_1 X_{it} + u_{it},$$

where  $\alpha_i$  are  $i$ -specific intercepts, referred as *fixed effects*.

Adding more variables of the two types, we get the Entity Fixed Effects Regression Model:

$$Y_{it} = \alpha_i + \beta_1 X_{1,it} + \beta_2 X_{2,it} + \dots + \beta_k X_{k,it} + u_{it}.$$

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# Regression with Panel Data: Time Fixed Effects Regression

Similarly, we can assume that

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 Z_t + u_{it},$$

where  $Z_t$  is an unobserved variable that depends on  $t$  but does not change with  $i$ . Therefore, we can rewrite

$$Y_{it} = \lambda_t + \beta_1 X_{it} + u_{it},$$

where  $\lambda_t$  are  $t$ -specific intercepts, referred as *time fixed effects*.

Adding more variables of the two types, we get the Time Fixed Effects Regression Model:

$$Y_{it} = \lambda_t + \beta_1 X_{1,it} + \beta_2 X_{2,it} + \dots + \beta_k X_{k,it} + u_{it}.$$

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# Regression with Panel Data: Both Entity and Time Fixed Effects Regression

Similarly, we can assume that there are both Entity and Time Fixed Effects:

$$Y_{it} = \alpha_i + \lambda_t + \beta_1 X_{1,it} + \beta_2 X_{2,it} + \dots + \beta_k X_{k,it} + u_{it}.$$